Distributed Continuous Time Algorithms for Communication-Free Regular Deployment of Homogeneous Multi-Agent Robotic Teams. ⋆

A.S. Matveev [a,b] P.A. Konovalov [a]

[a] *Mathematical Robotics Science Division, Sirius University, Olimpiyskiy ave. b.1, Sirius City, Krasnodar region, Russia (e-mail: almat1712@yahoo.com, petrkon98@mail.ru).*

[b] *Department of Mathematics and Mechanics, Saint Petersburg University, St. Petersburg, Russia.*

Abstract

A fully homogeneous team of an unknown size consists of identical agents that carry no communication facilities, are incapable of self-identification, cannot distinguish among the peers or play distinct roles and so should be subjected to a common control rule. On the positive side, every agent has access to the relative state of any peer that is within a finite "visibility range" provided that nothing obscures the view to the peer. Distributed continuous-time computationally inexpensive algorithms are offered for solution of the following problems: self-deployment of the team into nodes of a uniform lattice with a given step without gaps and multiple occupancy of nodes; reaching consensus on a quantity of interest (QoI); even self-distribution between two given values of QoI. These findings are illustrated by design of navigation strategies for, firstly, distributed communication-free cruise control of a platoon of vehicles on a road, secondly, even self-distribution of mobile sensors on a curvy barrier line, thirdly, navigation of planar robotic swarms for dense sweep coverage of corridor environments, and fourthly, autonomously building a virtual antenna array in 3D by mobile devices. The proposed algorithms are justified by mathematically rigorous global convergence results; their performance is confirmed by computer simulation tests.

Key words: Multiagent systems, generalized consensus, coordination, distributed algorithms, stability analysis

1 Introduction

The phenomenon of self-coordination and self-organization in large networks of interacting dynamical agents has recently attracted enormous interest in various research communities [1–3]. The primary focus is on comprehending the mechanism of emerging, in a spontaneous and decentralized fashion, an intelligent behavior of the entire team from local interactions among the teammates, each with a highly limited capacity and influence. The most intriguing is the case where the team is fully homogeneous, i.e., the agents are identical to one another, cannot distinguish among the peers, and cannot play distinct roles. The latter means that the agents should be individually driven by a common rule and their division into, e.g., decision-makers and executors, or leaders and followers, or "beacons" and "beacon users" is impossible. Such a homogeneity brings many practical benefits in terms of scalability, flexibility, fault tolerance, maintenance, repair, and ease of production.

Among the coordination problems, that of reaching consensus upon a quantity of interest (QoI) has attracted most attention [4–8]. This is partly due to the fundamental nature of the respective algorithms: by using and extending them, other coordination problems (e.g., those of formation [9] or robotic coverage [10] control) can be solved [4–8]. The wide variety of available distributed algorithms for consensus reaching includes those implementable on homogeneous teams; the latter solutions are illustrated by the nearest neighbors rule [11] with equal strengths of couplings around any given agent.

This paper addresses distributed self-organization of homogeneous teams in several scenarios that do not fully fit into the framework of the standard and well-researched consensus reaching task. In the first of them, the agents should line up into a 1D lattice with a pre-specified gap between the adjacent elements. So they should agree upon the position of the lattice and the assignment of its points to the agents so that all points are occupied

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and no point is inhabited twice. The team's size is unknown, no priorities or identities are accessible, e.g., in the form of the serial number in a certain enumeration. Any agent has access to data about a peer only if the latter in within a given "visibility range" and nobody obscures the view to it, and only in the form of the relative value of QoI. A paradigmatic example arises where QoI is a positional coordinate measured by sensors and the agents carry no communication facilities. Two another studied scenarios assume that an end-point of the lattice is given or, respectively, the step of the lattice is not stipulated and the agents should evenly distribute themselves between two given extreme values of QoI.

The above scenarios model certain aspects of a variety of real-life problems. For these problems, the algorithms developed in this paper can be used as building bricks to easily acquire completed and rigorously justified algorithmic solutions. This is illustrated for the following problems of self-interest: **1)** distributed communication-free cruise control of a platoon of vehicles on a curvy unknown road, **2)** even self-distribution of mobile sensors on a curvy barrier line, **3)** navigation of a robotic swarm for dense sweep coverage of corridor environments, **4)** arranging several UAV's into a 3D virtual antenna array.

To serve 3,4) and as a technically close theme, a new consensus protocol is proposed. It is special in that every agent uses only data from the most distant neighbors, contrary to the nearest neighbors rule. As is shown, this permits us to relax the standard connectivity requirements by redirecting them to a graph, which is more populated with links than those conventionally used in studies on distributed consensus. It should be also noted here that some form and degree of connectivity of the information flow graph is generally recognized as a pre-requisite for convergence of distributed consensus-reaching and coordination algorithms. This has motivated specialized algorithmic solutions for preserving the initial connectivity [12–19], which often considerably increase the overall computational burden. The algorithms proposed in this paper avoid the strain of connectivity preserving since they converge even if the above graph is disconnected initially. This is due to the inevitable birth of the connectivity when running the proposed computationally cheap algorithms.

For the problem of self-distribution between two values of QoI, the proposed algorithm has similarities with the virtual force (VF) approach to self-deployment of mobile sensor networks for area coverage [20–23]. This approach typically uses an interplay between virtual both repulsive and attractive forces and takes into account those exerted by all objects within a given range of distances, whereas our algorithm decreases the computation and communication loads by using only attraction and at most two closest objects. Many of the VF algorithms are burdened by elements of centralized decision-making [24], extended computational complexity [25], oscillation of nodes [26], access to global information [27], or reliance on clock synchronization. A common problem with these algorithms is that they are mainly heuristics-based and are not rigorously and completely justified.

On the contrary, the algorithms proposed in this paper are computationally cheap, directly convert the current observations into the current control, and operate in continuous time, which largely eliminates troubles of clock synchronization in contrast to discrete-time methods. The paper not only offers a mathematically rigorous justification of the convergence of these algorithms but also develops general techniques of the respective analysis. These algorithms and results can be used as building bricks for constructing control rules concerned with more complex problems and for proving their convergence. Theoretical findings of the paper are confirmed and complemented via computer simulation tests.

The body of the paper is organized as follows. Sec. 2 offers the problem setup. The main results are given in Sec. 3–5, with each of them addressing its own problem. Extensions and applications of these results are discussed in Sec. 6 and 7, respectively. Sec. 8 reports on computer simulations, and Sec. 9 offers conclusions. The proofs of the main results are given in the appendices.

Throughout the paper, $[k_1 : k_2]$ is the set of integers j such that $k_1 \leq j \leq k_2$; := means "is used to denote", "is defined to be"; $\|\vec{v}\|$ is the Euclidean norm of the vector $\vec{v}$.

2 Problem statement

A number of $N \geq 2$ agents, labeled 0 through $N-1$, operates in continuous time t. At any t a quantity $x_i(t) \in \mathbb{R}$ is assigned to agent i and called its *state*. Agent i affects evolution of the state x_i by generating the *control* $u_i(t)$:

$$\dot{x}_i = u_i, \qquad |u_i| \leq \overline{u}, \tag{1}$$

where $\overline{u} > 0$ is given. Agent i does not know the team's size N but is aware of the relative states $x_j(t) - x_i(t)$ of the peers j from a time-varying set $\mathcal{I}_i(t)$, which are called its *informants*. These data are anonymous: the label j is not recognizable. There is no hierarchy among the agents, and they are unable to play distinct roles.

A *protocol* is a common rule by executing which every agent individually builds its own control based on the available data. This rule should be such that **1)** for any initial state $\{x_i(0)\}_{i=0}^{N-1}$ of the team, it gives rise to a trajectory defined for all $t \geq 0$ and **2)** any permutation of $x_j(t)$ with $j \in \mathcal{I}_i(t)$ does not affect the output $u_i(t)$. Here 2) is motivated by the anonymousness of the data.

The task is to design protocols that ensure the following respective properties.

Defintion 2.1 *The agents are said to achieve*

1) *Consensus if there exists x_* such that $x_i(t) \to x_*$ as $t \to \infty$ for all agents $i \in [0 : N-1]$;*
2) *Consensus with a given anchor point $x_\dagger$ if the properties from* **1)** *hold with $x_* := x_\dagger$;*
3) *An even self-deployment with the step $w \geq 0$ if there exists x_* and an enumeration of the robots such that $x_i(t) \to x_* + iw$ as $t \to \infty$ for all agents $i \in [0 : N-1]$;*
4) *An even self-deployment with the step w and a given anchor point $x_\dagger$ if* **3)** *is met with $x_* := x_\dagger + w$;*
5) *An even self-deployment within an interval $[x_-, x_+]$ if there exists an enumeration of the agents such that $x_i(t) \to x_- + (i+1)\frac{x_+ - x_-}{N+1}$ $\forall i \in [0 : N-1]$ as $t \to \infty$.*

Here $w, x_-, x_+, x_\dagger$ are given; with $w = 0$, 3) and 4) are identical to 1) and 2), respectively. In 5), the limit values of N states x_i partition $[x_-, x_+]$ into $N+1$ equilength subintervals. In the context of either 2), or 4), or 5), the agents treat $x_\dagger$ and $x_\pm$ (if accessible) as the state of a "fictitious" agent and process them accordingly. If achieving the traits from 3)–5) is set as the goal, we assume that the status of being an informant of i is correlated with the distance to $x_i(t)$ as follows.

Assumption 1 *There is a constant $R_{\text{vis}} > 0$ such that agent $j \neq i$ is in $\mathcal{I}_i(t)$ whenever $|x_i(t) - x_j(t)| < R_{\text{vis}}$ and there is no agent whose state lies in the open interval with the end-points $x_i(t)$ and $x_j(t)$. Also, $i \notin \mathcal{I}_i(t)$.*

Example 1. Any agent gets data about the partners via observation of its neighborhood $\mathcal{N}$ within the sensing range R_{vis}. A partner from $\mathcal{N}$ is the informant only if the direct sight to it is not obstructed by a third agent.

Example 2. The agents inform one another about their states via communication channels, R_{vis} is the communication range. Intermediate agents may, but not necessarily, suppress signals from more remote peers.

If 4) in Defn. 2.1 is set as the objective, we assume that

4.a. If $|x_\dagger - x_i(t)| < R_{\text{vis}}$ and there is no agent whose state lies between $x_i(t)$ and $x_\dagger$, agent i has access to the relative "location" $x_\dagger - x_i(t)$ of the anchor;
4.b. Initially, the states of the agents are above $x_\dagger$.

In theory, the "symmetric" task of even self-deployment not above but below the anchor is reduced to 4) via the change of the variables $x_i \mapsto -x_i, u_i \mapsto -u_i, x_\dagger \mapsto -x_\dagger$.

If 5) is set as the control objective, we assume that

5.a. If $|x_\star - x_i(t)| < R_{\text{vis}}$ for an end $x_\star$ of the interval $[x_-, x_+]$ and there is no agent whose state is between $x_i(t)$ and $x_\star$, agent i has access to $x_\star - x_i(t)$;
5.b. Initially, the states of the agents are in (x_-, x_+).

We basically assume that any agent cannot differentiate between an agent and anchor/(interval's end), and equally treats the both as an "accessible object". Remarks on the opposite case are presented in 4) of Sec. 6.

Studies on distributed formation control expose the need to avoid the situations where the team splits into parts that get no information about each other. Now this concern is about the connectivity of the graph $\Gamma_{\text{all}}(t)$ with the set of nodes $[0 : N-1]$ where an edge goes from j to i if and only if $j \in \mathcal{I}_i(t)$. Along with $\Gamma_{\text{all}}(t)$, we consider its undirected subgraph $\Gamma(t)$ with the same nodes in which i and j are linked if and only if $i \neq j$, $|x_i(t) - x_j(t)| < R_{\text{vis}}$, and no agent lies between $x_i(t)$ and $x_j(t)$.

Defintion 2.2 *A state $\{x_i\}_{i=0}^{N-1}$ of the team is said to be*

- non-breaking *if the related graph Γ is connected;*
- collisionless *if $x_i \neq x_j \forall i \neq j$, and $x_i > x_\dagger \forall i$ in the context of* **4)** *in Defn.* 2.1 *and $x_- < x_i < x_+ \forall i$ in the case of* **5)**.

A protocol is said to be

a) collisionless *if the state is collisionless for all $t \geq 0$ whenever it is collisionless initially;*
b) connectivity saving *if the graph $\Gamma(t)$ does not loose edges as time progresses.*

It is assumed that the properties from **a)** and **b)** should hold in closed-loop if the protocol at hands is applied.

The above ban on dropouts of the links in the subgraph $\Gamma(t)$ of $\Gamma_{\text{all}}(t)$ is a weaker requirement than the popular ban on loss of edges in $\Gamma_{\text{all}}(t)$ [12, 13, 16, 19]. Moreover, the latter ban may be overly restrictive. For example, it disallows the quantity $|x_i(t) - x_j(t)|$ to cross R_{vis} from below up in example 2 (with no signal suppression) and thereby, makes 3)–5) from Defn. 2.1 unattainable if $|x_i(0) - x_j(0)| < R_{\text{vis}}$ $\forall i \neq j$ but the distance between the extreme states in the targeted deployment exceeds R_{vis}. Meanwhile, the ban restricted to $\Gamma(t)$ is compatible with the attainability of 3)–5), as will be shown later on.

There also are methods of connectivity preservation that let $\Gamma_{\text{all}}(t)$ to loose edges [14, 17, 18]. However, they are computationally expensive and typically rely on extended either communication among the agents or their awareness about the graph topology, unlike the methods proposed in this paper.

For collisionless protocol and initial state, $x_i(t) \neq x_j(t)$ $\forall t \geq 0, i \neq j$. So the agents can be enumerated so that

$$x_0(t) < x_1(t) < \ldots < x_{N-1}(t) \qquad \forall t. \tag{2}$$

The properties from Defn. 2.1 are hardly attainable if the agents cannot obtain information about one another whenever the targeted property is nearly achieved. This remark motivates the following respective requirements in the cases 3), 4), and 5) from Defn. 2.1:

$$w < R_{\text{vis}}, \qquad (x_+ - x_-)/(N+1) < R_{\text{vis}}. \qquad (3)$$

3 Even self-deployment with a given step

We aim at 3) from Defn. 2.1 and put $\Delta x_{i,j} := x_j - x_i$.

Protocol 1: At time t, agent i carries out the following:

s.1) Finds the set of *essential upper/lower informants*

$$\mathcal{I}_i^{\pm}(t) := \left\{ j \in \mathcal{I}_i(t) : R_{\text{vis}} > \pm \Delta x_{i,j}(t) > 0 \right\}; \quad (4)$$

s.2) Finds the gaps with the upper and lower informants if they do exist, and takes w as this gap otherwise:

$$d_i^{\pm}(t) := \begin{cases} \min\limits_{j \in \mathcal{I}_i^{\pm}(t)} |\Delta x_{i,j}(t)| & \text{if } \mathcal{I}_i^{\pm}(t) \neq \emptyset, \\ w & \text{otherwise;} \end{cases} \quad (5)$$

s.3) Computes the current control input:

$$u_i(t) := \Xi[d_i^+(t)] - \Xi[d_i^-(t)]. \quad (6)$$

Here the smooth map $\Xi(\cdot) : [0, \infty) \to [0, \overline{u}]$ is to be pre-selected by the designer of the controller so that

$$\Xi'(d) > 0 \quad \forall d \in [0, R_{\text{vis}}), \qquad \Xi(0) = 0. \quad (7)$$

Then the control (6) meets the inequality from (1).

Remark 3.1 The protocol is conformable to the anonymousness of the data since re-enumeration of the agents does not affect the quantities (5) and so the control (6).

The solution of the discontinuous closed-loop system is meant in the Fillipov's sense [28]. Given an initial state, this solution exists for all $t \geq 0$ since the controls are bounded. Uniqueness of the solution is a more intricate issue and to meet the paper length limit, we omit its discussion and consider all solutions.

Theorem 3.1 *Let Asm.* 1*, the first part of* (3)*, and* (7) *hold. The following statements are true for the protocol* 1*:*

i) *The protocol is connectivity saving and collisionless;*
ii) *If the initial state is non-breaking and collisionless, the agents achieve an even self-deployment with the step w (as is defined in* **3)** *of Defn.* 2.1*).*

The proof of this theorem is given in App. A.

In the case from **ii)**, the team avoids the discontinuities of the control law by **i)**. Indeed, due to (4)—(6), they hold only at states where either $x_i = x_j$ or $x_i \pm R_{\text{vis}} = x_j$ for some $i \neq j$, which states are not visited due to **i)**.

4 Achieving consensus

By Thm. 3.1, this is achieved by protocol 1 with $w = 0$ under the assumptions from ii). Meanwhile, they can be relaxed. We show this for the modified *protocol* $1_{\max}$ in which $w := 0$ and min is replaced by max in (5), and $R_{\text{vis}} := \infty$ in (4), (7), which potentially enlarges $\mathcal{I}_i^{\pm}(t)$.

In the context of 2) in Defn. 2.1, we assume that N and the enumeration of the agents take account of the anchor, which is labeled by 0. If a true agent i has access to the relative state $x_\dagger - x_i$ of the anchor, 0 is included in $\mathcal{I}_i$ and i is put in $\mathcal{I}_0$, though $\mathcal{I}_0$ is not used for control generation. Then 0 is well-defined as a node of $\Gamma_{\text{all}}(t)$.

For simplicity, now we focus on the case of bilateral data exchange: $j \in \mathcal{I}_i(t) \Leftrightarrow i \in \mathcal{I}_j(t)$. The *graph of essential links* Γ_{ess} is that with the nodes $0, \ldots, N-1$, where i and j are linked if and only if the Lebesgue measure of the set $\left\{ t \geq 0 : i \,\text{and}\, j \text{ are linked in } \Gamma_{\text{all}}(t) \right\}$ is infinite.

Theorem 4.1 *Let* (7) *hold and let $j \in \mathcal{I}_i(t) \Leftrightarrow i \in \mathcal{I}_j(t) \forall t \geq 0$. Then the following claims are true for the protocol* $1_{\max}$ *in the context of both* 1) *and* 2) *of Defn.* 2.1*:*

i) *The range $\mathcal{R}(t) := \max_{i,j} |\Delta x_{i,j}(t)|$ does not grow up;*
ii) *For any sign, $\max_i \pm x_i(t)$ converges to a finite limit as $t \to \infty$ and does not increase;*
iii) *If Γ_{ess} is connected, the consensus is achieved, which is with the anchor $x_\dagger$ in the context of* 2) *in Defn.* 2.1.

The proof of this theorem is given in App. B.

If $j \in \mathcal{I}_i(t) \Rightarrow |\Delta x_{i,j}(t)| \leq R_*$, the behavior of the map Ξ outside $[0, R_*]$ does not affect u_i by (4)–(6). So the first requirement from (7) can be confined to $d \in [0, R_*]$. By i) in Thm. 4.1, these are true with $R_* := \mathcal{R}(0)$.

For some applied problems, Sec. 7.3 and 7.4 prove that the protocol $1_{\max}$ (coupled with a protocol from Sec. 5) guarantees consensus even if $\Gamma_{\text{all}}(t)$ is initially disconnected. Most consensus-reaching algorithms either rely on maintenance of the initial connectivity or merely assume the needed connectivity on $[0, \infty)$, with leaving the reasons for the presence of this property an open issue.

5 Even self-deployment with a given anchor and within a given interval, respectively.

Protocol 2 is to be used if an anchor $x_\dagger$ is given, and offers every agent to run protocol 1, with routinely interpreting $x_\dagger$ as the state of a teammate and putting R_{vis} in place of w in the second row of (5) if $\mathcal{I}_i^-(t) = \emptyset$.

Theorem 5.1 *Let Asm.* 1*, the first part of* (3)*, and* (7) *be true. Then for the protocol* 2*,* **i)** *in Thm.* 3.1 *holds and if the initial state is collisionless, the agents achieve an even self-deployment with the step w and the anchor $x_\dagger$.*

The proofs of the results from this section are given in App. C. The conclusion of Thm. 5.1 holds even if initially the states of the agents are arbitrarily far from the anchor and one another and the information graph $\Gamma_{\text{all}}(0)$ is fully disconnected (edge-less). By (3), Asm. 1, and Thm. 5.1, the graph eventually becomes connected.

Protocol 3 is used to achieve **5)** in Defn. 2.1. It orders any agent to run protocol 1 with a special $w > 0$, while routinely handling x_- and x_+ as the states of agents.

Theorem 5.2 *Let Asm.* 1*, the second part of* (3)*, and* (7) *hold. Suppose that the parameter w of the protocol* 3 *is chosen so that $(x_+ - x_-)/(N+1) < w < R_{\text{vis}}$. Then* **i)** *in Thm.* 3.1 *is true, and if the initial state is collisionless, the agents achieve an even self-deployment within $[x_-, x_+]$.*

The delineated choice of w is feasible due to the second part of (3). The conclusion of Thm. 5.2 holds even if initially, the gaps between some pairs of adjacent agents (including the fictitious ones $x_\pm$) exceed R_{vis} and so the graph $\Gamma_{\text{all}}(0)$ is disconnected.

In the context of Thm. 5.2, the agents do not need to know the length $l := x_+ - x_-$ of the interval and the team's size N to run the protocol. To tune it, knowledge of an upper $l_+ \geq l$ and lower $N_- \leq N$ bounds is enough: it suffices to pick w so that $l_+/N_- < w < R_{\text{vis}}$. This is feasible if $l_+/N_- < R_{\text{vis}}$, which condition is intelligible: it means that within the available knowledge, there are no cases where the goal is unreachable since the agents lose mutual visibility when being deployed as is desired.

Rem. 3.1 remains true for the protocols $1_{\max}$, 2, and 3.

6 Some other scenarios

1. Let in the context of Thm. 3.1, 5.1, or 5.2, the initial state be collisionless and, in the case of Thm. 3.1, nonbreaking. Suppose that sometimes the team looses members, while continuing to run the protocol "as usual" with ignoring the lost agents. By consecutively shifting the origin of the time axis to the moments of losses and applying the respective theorem, we see that its conclusion remains valid under the following reservations:

- **r.1)** For Thm. 3.1: *whenever an agent j disappears at some time t, any remaining agent i should have a remaining peer $k \neq i$ such that $|\Delta x_{i,k}(t)| < R_{\text{vis}}$ and $\mathbf{sgn}\,\Delta x_{i,k}(t) = \mathbf{sgn}\,\Delta x_{i,j}(t)$;*
- **r.2)** For Thm. 5.2: *the second part of* (3) *should hold for the eventual size N_* of the team $(x_+ - x_-)/(N_* + 1) < R_{\text{vis}}$ and the parameter w should meet the inequalities $(x_+ - x_-)/(N_* + 1) < w < R_{\text{vis}}$.*

By **r.1)**, the connectivity of $\Gamma(t)$ is not violated. No reservations are imposed in the case of Thm. 5.1.

2. Let extra team members sometimes enter the scene, after which the entire united team runs the protocol "as usual". Suppose also that whenever a newcomer appears, its state is different from those of all other agents. By consecutively shifting the origin of the time axis to the moments of admissions and applying the respective theorem, we see that its conclusion remains valid.

3. Let the agents be able to distinguish between a peer and the anchor or interval's end. Then in (5), any agent can replace the true distance d to the anchor or that end by the scaled distance αd with a given $\alpha > 0$. An inspection of the proofs in App. C shows that then Thm. 5.1, 5.2 basically remain true. Alterations are as follows.

For Thm. 5.1: The first part of (3) is enriched with $\alpha^{-1}w < R_{\text{vis}}$; in 4) of Defn. 2.1, the asymptotics of x_i's change into $x_i(t) \to x_\dagger + w/\alpha + iw \ \forall i \in [0 : N-1]$ as $t \to \infty$. Thus, the distance between any two adjacent true agents goes to w, as before, whereas the distance from the "lowest" of them to the anchor goes to w/α.

For Thm. 5.2: The second part of (3) is replaced by $\max\{1, \alpha^{-1}\}\Delta < R_{\text{vis}}$; the asymptotics of x_i's change into $x_i(t) \to x_- + \alpha^{-1}\Delta + i\Delta \ \forall i \in [0 : N-1]$ as $t \to \infty$, where $\Delta := (x_+ - x_-)/(2\alpha^{-1} + N - 1)$.

Now we offer two examples involving scaled distances.

i) Mobile sensors with a sensing range R_{sen} must allocate themselves above an anchor A so that any point from the widest possible interval rooted at A be shadowed by a sensor. This means that a sensor is at a distance of R_{sen} above A, and the others are evenly distributed up higher with the step $w = 2R_{\text{sen}}$. In this case, $\alpha = 2$.

ii) The agents must form the densest net across $X := [x_-, x_+]$. This net minimizes $\max_{x \in X} \min_i |x - x_i|$ over all deployments of x_i within X. As can be shown, the optimal deployment is such that the distances Δ_x between the adjacent agents are equal to one another, whereas the distance from any end of the interval to the closest state x_i is $1/2\Delta_x$. In this case, $\alpha = 2$, as before.

7 Illustrative examples.

7.1 Distributed communication-free cruise control

A platoon of N robotic vehicles, labeled by $i \in [0 : N-1]$, move, one by one, along a curvy road. A common spacing w between the vehicles and speed $v_\rightarrow \geq 0$ are recommended. The vehicles are laterally stabilized so that the position of robot i is characterized by its distance x_i along the road from a fixed point. The robots are not equipped with communication facilities. On-board sensors inform any robot of, at least, the road distance to the closest vehicle both in front and behind if it exists and that distance does not exceed a given threshold $R_{\rm vis}$. A common control rule should be designed that enables the platoon to autonomously achieve the desired speed and spacing, when being independently run by every robot.

Robot i is described by $\dot{x}_i = u_i, |u_i| \leq \overline{v}$, where the speed u_i is the control input and $\overline{v}$ is given. We assume that the speed $v_\to$ is feasible $v_\to < \overline{v}$ and the robots do not lose the sensorial contact when being close to the targeted inter-vehicle spacing $w < R_{\rm vis}$.

The solution is given by protocol 1, where (6) is altered into $u_i(t) := v_\to + \Xi[d_i^+(t)] - \Xi[d_i^-(t)], \overline{u} := \overline{v} - v_\to$ in (7), and in (4), $\mathcal{I}_i(t)$ consists of the robots about which robot i gets sensory data. Indeed, let the initial positions be different and such that the distance between the adjacent locations is less than $R_{\rm vis}$. By applying Thm. 3.1 to $z_i(t) := x_i(t) - v_\to t$, we see that the vehicles do not collide with or overtake one another, the inter-vehicle distances go to w, and their speeds go to $v_\to$ as $t \to \infty$: since $d_i^\pm(t) \to w$, we have $u_i(t) \to v_\to + \Xi[w] - \Xi[w] = v_\to$.

The above findings extend to the case where $v_\to = v_\to(t)$ if every robot has access to the speed profile $v_\to(\cdot)$.

7.2 Distribution of mobile devices over a curvy barrier

The same robots must reach steady positions evenly distributed over the part of the road between two marks x_- and x_+. Any robot i can recognize the mark and measure the road distance to it when being close enough: $|x_i - x_\pm| < R_{\rm vis}$. The robots start with different positions within (x_-, x_+) and do not lose the sensorial contact when being close to the even distribution $\frac{x_+ - x_-}{N+1} < R_{\rm vis}$.

By Thm. 5.2, this mission is solved by protocol 3 with the parameter w chosen so that $\frac{x_+ - x_-}{N+1} < w < R_{\rm vis}$.

If the curve is a meander, this solution can be used to uniformly fill an area or volume. We leave elaboration of the respective details as a topic of further research.

7.3 Dense sweep coverage of corridor environments.

A swarm of N planar robots operates in a *corridor* $\mathfrak{C}$ between two parallel lines. The swarm must arrange itself into a straight uniform lattice that evenly fills the corridor width, is perpendicular to $\mathfrak{C}$, and moves along $\mathfrak{C}$ with the given speed $v_\to > 0$ as a whole unit. Collisions among the robots and with the corridor sides are prohibited. The robots are not equipped with communication facilities, do not discern the sensed objects, and are unaware of the team's size N and the corridor width W. In its local frame, any robot has access to the corridor direction and to the relative locations of the objects, including the walls, within a given sensing range $R_{\rm sen} > 0$.

We consider robots with the simple integrator kinematics that are able to travel at speeds not greater than $\overline{v}$:

$$\dot{\boldsymbol{r}}_i = \boldsymbol{v}_i, \quad \|\boldsymbol{v}_i\| \leq \overline{v}. \tag{8}$$

Here $\boldsymbol{r}_i(t) \in \mathbb{R}^2$ is the location of robot i, the velocity vector $\boldsymbol{v}_i = \boldsymbol{v}_i(t)$ is the control input, and $\overline{v} > v_\to$.

We introduce a Cartesian frame for which the corridor sides are described by $y = \pm W/2$ and the x-axis goes in the desired direction of motion. Let $\chi_i(t)$ and $y_i(t)$ stand for abscissa and ordinate of $\boldsymbol{r}_i(t)$. We "split" $\Delta_v := \overline{v} - v_\to > 0$ and $R_{\rm sen}$ between the x- and y-axes by picking $\Delta_v^x, \Delta_v^y > 0$ and $R_{\rm vis}^x, R_{\rm vis}^y > 0$ such that $(\Delta_v^x)^2 + (\Delta_v^y)^2 \leq \Delta_v^2$ and $(R_{\rm vis}^x)^2 + (R_{\rm vis}^y)^2 \leq R_{\rm sen}^2$. Finally, we pick two maps $d \in [0,\infty) \mapsto \Xi_x(d) \in [0, \Delta_v^x], \Xi_y(d) \in [0, \Delta_v^y]$ that meet (7) with $R_{\rm vis} := R_{\rm vis}^x$ and $R_{\rm vis} := R_{\rm vis}^y$, respectively.

Protocol 4: At time t, agent i carries out the following:

1) Applies the protocol $1_{\max}$ with $\Xi := \Xi_x$ to the relative abscissas $\chi_j(t) - \chi_i(t)$ of all visible peers j to form $v_i^x(t)$;

2) Applies the protocol 3 with $\Xi := \Xi_y, R_{\rm vis} := R_{\rm vis}^y, x_\pm := \pm W/2$ to the relative ordinates $y_j(t) - y_i(t)$ of the visible objects to form $v_i^y(t)$;

3) Takes $\boldsymbol{v}_i(t) := [v_\to + v_i^x(t); v_i^y(t)]$ in (8).

The inequality from (8) is met since due to (6) and (7)

$$\begin{aligned} |v_i| &\leq v_\to + \sqrt{(v_i^x)^2 + (v_i^y)^2} \\ &\leq v_\to + \sqrt{(\Delta_v^x)^2 + (\Delta_v^y)^2} \leq v_\to + \Delta_v = \overline{v}. \end{aligned} \tag{9}$$

Proposition 1 *Let $R_{\rm vis}^y > \frac{W}{N+1}$ and initially the robots be placed inside the corridor $|y_i(0)| < W/2$ so that $\varsigma := \max_{i,j} |\chi_i(0) - \chi_j(0)| < R_{\rm vis}^x$ and $y_i(0) \neq y_j(0) \forall i \neq j$. Then the objective is attained: the robots do not collide with one another and the corridor walls, the ordinates $y_i(t)$ achieve an even deployment within $[-W/2, W/2]$ (see Defn. 2.1), and there exists χ_* such that $\chi_i(t) - \chi_* - v_\to t \xrightarrow{t\to\infty} 0$.*

PROOF. Now $\mathcal{I}_i(t) = \{j \neq i : \|\boldsymbol{r}_j(t) - \boldsymbol{r}_i(t)\| < R_{\rm sen}\}$. Applying i) of Thm 4.1 to $z_i(t) := \chi_i(t) - v_\to t$ yields that $|\chi_i(t) - \chi_j(t)| = |z_i(t) - z_j(t)| \leq \varsigma < R_{\rm vis}^x \ \forall i,j$. Hence $|y_i(t) - y_j(t)| < R_{\rm vis}^y \Rightarrow \|\boldsymbol{r}_j(t) - \boldsymbol{r}_i(t)\| < \sqrt{(R_{\rm vis}^x)^2 + (R_{\rm vis}^y)^2} \leq R_{\rm sen} \Rightarrow j \in \mathcal{I}_i(t)$ if $j \neq i$, i.e. Asm. 1 (with $R_{\rm vis} := R_{\rm vis}^y$) holds for $y_i(t)$'s. By applying Thm. 5.2 to them, we arrive at the first two sentences in the conclusion. Then by 5) in Defn. 2.1, there is t_* and an enumeration of the robots such that for all $t \geq t_*$ and i, the gap $y_{i+1}(t) - y_i(t)$ is close to $W/(N+1)$ and so is less than $R_{\rm vis}^y$. By the foregoing, this implies that $i+1 \in \mathcal{I}_i(t)$. So $i+1$ and i are linked with an edge in

$\Gamma_{\text{all}}(t)$ and, since this holds for all $t \geq t_*$, in Γ_{ess}. Thus the graph Γ_{ess} is connected. The proof is completed by applying iii) in Thm. 4.1 and 1) in Defn. 2.1 to $z_i(t)$'s.

The assumption $R^y_{\text{vis}} > \frac{W}{N+1}$ of Prop. 1 is met for $R^x_{\text{vis}} \approx 0$, $R^y_{\text{vis}} \approx R_{\text{sen}}, R^y_{\text{vis}} < R_{\text{sen}}$ if $R_{\text{sen}} > \frac{W}{N+1}$. The last claim means that the sensorial touch among the robots is not lost near the desired deployment. Depending on R^x_{vis}, the assumption $\varsigma < R^x_{\text{vis}}$ does not necessarily mean that ς is small. It is small if, e.g., the robots are initially put to nearly the same place, which is fairly practical. Almost surely, $y_i(0) \neq y_j(0) \forall i \neq j$ if the robots are disembarked according to a continuous probability distribution.

Prop. 1 allows that some initial inter-robot y-gaps exceed R_{vis} and so the graph $\Gamma_{\text{all}}(0)$ is not connected.

7.4 Building a virtual antenna array in 3D

The robots from Sec. 7.3 operate in 3D above a horizontal plane H. Any of them carries a transceiver. It is needed to vertically line the robots up above a certain "beacon" point $\boldsymbol{p} \in H$ so that the distance gap between the adjacent robots equals w and the altitude of the lowest of them equals $h_\dagger$, where $w > 0$ and $h_\dagger > 0$ are given.

We introduce the (absolute) Cartesian frame (AF) rooted at $\boldsymbol{p}$, with the z-axis going perpendicularly to H and upwards. In its local frame, any robot i has access to the coordinate directions of AF and to the relative coordinates of $\boldsymbol{p}$ if $\|\boldsymbol{r}_i - \boldsymbol{p}\| < R_{\text{sen}}$. These enable robot i to compute its own altitude h_i. The robots do not lose the sensorial contact with one another and H when being close to the targeted distribution: $R_{\text{sen}} > w, h_\dagger$.

We "split" $\overline{v}$ and R_{sen} between the axes by choosing $\overline{v}^b, R^b_{\text{vis}} > 0$ for $b = x, y, z$ so that $\sum_b (\overline{v}^b)^2 \leq \overline{v}^2$ and $\sum_b (R^x_{\text{vis}})^2 \leq R^2_{\text{sen}}, R^z_{\text{vis}} > w, h_\dagger$; this is feasible since $R_{\text{sen}} > w, h_\dagger$. Also, we pick a function Ξ_b that maps $[0, \infty)$ to $[0, \overline{v}^b]$ and meets (7) with $R_{\text{vis}} := R^b_{\text{vis}}$.

Protocol 5: At time t, agent i carries out the following:

1) For $b := x, y$, applies the protocol $1_{\max}$ with $\Xi := \Xi_b$ to the relative coordinates $b_j(t) - b_i(t)$ of all visible objects j (including $\boldsymbol{p}$, if applicable) to form $v^b_i(t)$;

2) Applies the protocol 2 with $\Xi(\cdot) := \Xi_z(\cdot), R_{\text{vis}} := R^z_{\text{vis}}$, the anchor $x_\dagger := h_\dagger$, and step w to the relative applicates $z_j(t) - z_i(t)$ of those objects to form $v^z_i(t)$;

3) Take $\boldsymbol{v}_i(t) := [v^x_i(t); v^y_i(t); v^z_i(t)]$ in (8).

Similarly to (9), this output $\boldsymbol{v}_i(t)$ meets (8).

Proposition 2 *Let the robots be initially disembarked at different altitudes $h_i(0) > h_\dagger$ above the rectangle given by $|x| < R^x_{\text{vis}}, |y| < R^y_{\text{vis}}, z = 0$. Then the goal is attained: the robots do not collide with one another and H, their projections onto H go to $\boldsymbol{p}$ as $t \to \infty$, and $z_i(t) \to h_\dagger + iw \ \forall i$ under proper enumeration of the robots.*

PROOF. We put $\boldsymbol{r}_{-1}(t) := \boldsymbol{p}$ and interpret x_{-1}, y_{-1}, z_{-1} accordingly. Similarly to the opening part of the proof of Prop. 1, we infer that $|x_i(t)| < R^x_{\text{vis}}, |y_i(t)| < R^y_{\text{vis}} \ \forall t \geq 0, i$ and so $|z_i(t) - z_j(t)| < R^z_{\text{vis}} \Rightarrow \|\boldsymbol{r}_j(t) - \boldsymbol{r}_i(t)\| < R_{\text{sen}} \Rightarrow j \in \mathcal{I}_i(t)$ if $j \neq i$, i.e. Asm. 1 holds. By Thm. 5.1 and Defn. 2.1, 2.2, $z_i(t) \neq z_j(t) \ \forall i \neq j, t \geq 0$ and so the robots do not collide with one another and H. Also, they can be enumerated so that $z_i(t) \to h_\dagger + iw \ \forall i$ as $t \to \infty$. Like in the proof of Prop. 1, this implies that for all $i \in [-1 : N-2]$, the nodes $i+1$ and i are linked with an edge in $\Gamma_{\text{all}}(t)$ for $t \approx \infty$ and so in Γ_{ess}. Thus the graph Γ_{ess} is connected. The proof is completed by consecutively applying iii) in Thm. 4.1 and 2) in Defn. 2.1 first, to $x_i(t)$'s and $x_\dagger := 0$ and second, to $y_i(t)$'s and $x_\dagger := 0$.

In Prop. 2, the initial distances between the robots along the z-axis may be arbitrarily large so that the information flow graph $\Gamma_{\text{all}}(0)$ is edge-less.

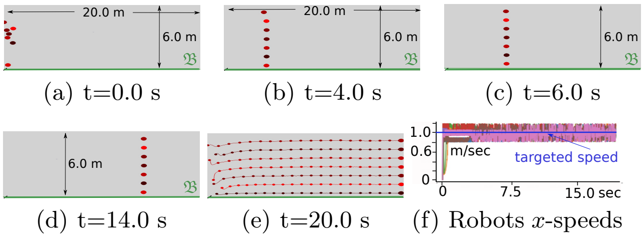
8 Computer Simulation Tests

The measurements of the relative locations were corrupted with i.i.d. noises uniformly distributed from -5.0 to 5.0 cm, the control update period is 20 ms, and $\overline{v} = 4.0$ m/s. In (6), $\Xi_x(d) = \Xi_y(d) = 4.0d$. The multimedia of all experiments are available at `https://shorturl.at/osuS3`.

The scenario in Fig. 1 is akin to that in Sec. 7.3. However, a corridor side is far away from the other one $\mathfrak{B}$ (or is absent). Seven robots with $R_{\text{vis}} = 0.9$ m should line up into a straight lattice that is perpendicular to $\mathfrak{B}$ and goes with the speed $v_\to = 1.0m/s$ along $\mathfrak{B}$. Any robot has a certain processing/monitoring range with the y-width $r = 0.35$ m. It is needed to distribute the robots so that the maximal "vertical" interval rooted at $\mathfrak{B}$ be covered by these ranges. This means that the "lowest" robot goes at a distance r from $\mathfrak{B}$, and the other robots maintain the vertical distance $2r$ to the nearest neighbors.

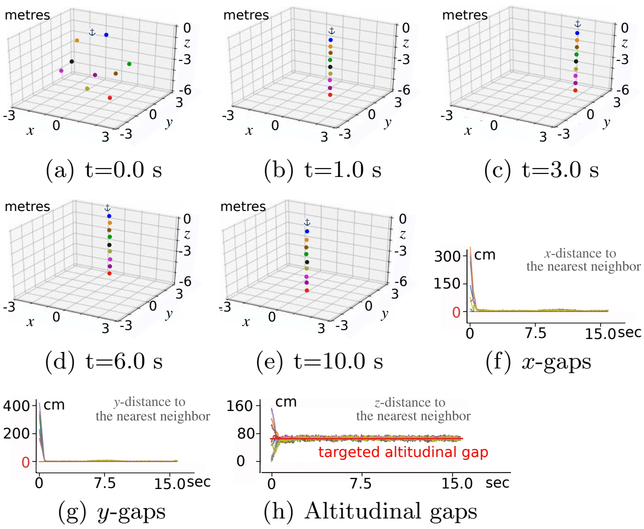
Protocol 4 from Sec. 7.3 is applied. However, its step 2) now uses protocol 2 from Sec. 5 (instead of protocol 3), where $w := 2r$ and $x_\dagger$ is the ordinate of $\mathfrak{B}$. Also we assume that the robots can recognize the side $\mathfrak{B}$ and, based on this, they employ the scaled y-distance to $\mathfrak{B}$ with $\alpha = 2$, as is described in 3) from Sec. 6.

By Fig. 1, the team achieves the targeted formation and x-speed with a good precision as early as at $t = 2.0$ s, starting out with an irregular deployment. Then the



(a) t=0.0 s (b) t=4.0 s (c) t=6.0 s (d) t=14.0 s (e) t=20.0 s (f) Robots x-speeds

Fig. 1. Sweeping the maximal strip near a straight landmark.



(a) t=0.0 s (b) t=1.0 s (c) t=3.0 s (d) t=6.0 s (e) t=10.0 s (f) x-gaps (g) y-gaps (h) Altitudinal gaps

Fig. 2. A virtual antenna array in 3D below a moving beacon.

robots stably maintain those speed and formation with the maximal speed error $\approx 0.2m/s$ and position error $\approx 4.0cm$. The latter error is more than condonable in the face of the sensor noise level $\approx 5.0cm$. Thus the objective is achieved with a good accuracy despite sensor errors.

Fig. 2 addresses the scenario from Sec. 7.4 with $N = 9$, $w = h_\dagger = 0.7$ m and the following alterations: the beacon ⚓ is not steady but moves in the plane $z = 0$ over a circle with a diameter of 3.0 m, the antenna is built not above but below ⚓, the sensing range 1.0 m along the z-axis differs from that 10.0m along the other axes. By Fig. 2(a), the team (including ⚓) initially consists of two clusters (three upper teammates versus the others), which are invisible for each other. Fig. 2(b-h) shows that driven by protocol 5, the team lines up in a straight line beneath the beacon and achieves the even inter-agent distribution with good accuracy as early as at ≈ 1.0 s. From this time forth, the algorithm stably maintains the required 3D antenna array configuration with the position error $\lesssim 6.0cm$ (comparable with the sensor errors). Thus the algorithm copes with the mission despite the initial disconnectivity of the information flow graph and continual displacement of the beacon.

9 Conclusions and future work

Distributed continuous-time computationally cheap algorithms were offered that ensure, respectively, self-deployment of the team into nodes of a uniform lattice with a given step; reaching consensus on a quantity of interest (QoI); even self-distribution between two given values of QoI. The performance of the algorithms was justified by mathematically rigorous global convergence results and was confirmed by computer simulation tests.

Networks of first-order agents have earned a reputation of instructive models for basic study of consensus dynamics and the likes of them. Many high-order consensus algorithms either are ideological heirs of first-order parents [4, 29] or are even formally based on them [30]. Our future work includes extension of findings of this paper to more complex models of the agents dynamics.

The background study of this paper deals with a scalar QoI. Meanwhile, coordinate-wise combination of the proposed algorithms enables one to handle some vector cases, as was shown in Sec. 7.3 and 7.4. This approach has a potential of building some popular higher-dimensional formations (like an uniform coverage of a rectangle, or parallelepiped, or a more complex geometric figure) via using curvilinear coordinates related to tricky curves (e.g., a meander), as is sketched out at the end of Sec. 7.2. Elaboration of these issues, as well as direct extension of the basic algorithms on higher dimensions, are also on our agenda.

A Proof of Theorem 3.1.

In the appendices, "a.a." is an abbreviation for "almost all". We study a trajectory of the system driven by the protocol 1.

Lemma A.1 $x_i(0) < x_j(0) \Rightarrow x_i(t) < x_j(t)\ \forall t \geq 0$.

PROOF. We put $E := \{(i,j) : x_i(0) < x_j(0)\}$. Let $[0, \tau)$ be the leftmost connected component of the open (in $[0, \infty)$) set $\{t \geq 0 : x_i(t) < x_j(t)\ \forall (i,j) \in E\} \ni 0$. Suppose that the lemma fails to be true. Then $\tau < \infty$ and there are $G \subset [0 : N-1]$ and $x_* \in \mathbb{R}$ such that $x_i(\tau) = x_*\ \ \forall i \in G, x_i(\tau) \neq x_*\ \forall i \notin G$ and $i', i'' \in G$ for some $(i', i'') \in E$. Let $m_\pm(t) := \pm \max_{i \in G} \pm x_i(t)$ and let $i^\pm(t)$ be a related maximizer. Then $\Delta(t) := m_+(t) - m_-(t) > 0\ \forall t \in [0, \tau)$ and $\Delta(\tau) = 0$. By the Danskin's theorem [31], the function $m_\pm(\cdot)$ is absolutely continuous and $\dot m_\pm(t) = \dot x_{i^\pm(t)}(t)$ a.a. $t \in [0, \tau)$. So for a.a. $t < \tau, t \approx \tau$, we have

$$\dot\Delta(t) = \dot m_+(t) - \dot m_-(t) = \dot x_{i^+(t)}(t) - \dot x_{i^-(t)}(t)$$
$$\overset{(1)}{=} u_{i^+(t)}(t) - u_{i^-(t)}(t) \overset{(6)}{=} \Xi[\underbrace{d^+_{i^+(t)}(t)}_{\geq 0}] - \Xi[\underbrace{d^-_{i^+(t)}(t)}_{\leq \Delta(t) \text{ by (4), (5)}}]$$
$$-\Xi[\underbrace{d^+_{i^-(t)}(t)}_{\leq \Delta(t) \text{ by (4), (5)}}] + \Xi[\underbrace{d^-_{i^-(t)}(t)}_{\geq 0}] \overset{(7)}{\geq} -2\Xi[\Delta(t)]. \qquad \text{(A.1)}$$

By (7), $\Delta_* \equiv 0$ solves the associated ODE $\dot{\Delta} = -2\Xi[\Delta]$, and $\Delta(\tau - \varepsilon) > \Delta_*(\tau - \varepsilon)$ if $\varepsilon > 0, \varepsilon \approx 0$. Then by [32, Ch. III, Th. 4.1], $\Delta(\tau) > \Delta_*(\tau)$, in violation of the above property $\Delta(\tau) = 0$. This contradiction completes the proof.

Corollary 1 *For any agents i, j, either $E_{i,j} := \{t \geq 0 : x_i(t) < x_j(t)\} = \emptyset$, or $E_{i,j} = [0, \infty)$, or $E_{i,j} = (\tau, \infty)$ with some $\tau > 0$, where $x_i(t) = x_j(t)\ \forall t \in [0, \tau)$ in the last case.*

Now we focus on the case where the following holds.

Assumption A.1 *The initial state is collisionless.*

We enumerate the agents in increasing order of $x_i(0)$ and put

$$\Delta_i(t) := x_{i+1}(t) - x_i(t),\ i \in [0 : N-2], \qquad \Delta_{-1}(t) := \Delta_{N-1}(t) := w. \quad \text{(A.2)}$$

Corollary 2 *Let Asm.* A.1 *be true. Then* (2) *holds.*

Corollary 3 *The protocol* 1 *is collisionless.*

Lemma A.2 *The protocol* 1 *is connectivity saving.*

PROOF. Due to the time-invariance of the protocol, Defn. 2.2, the definition of the graph $\Gamma(t)$, and Cor. 1, it suffices to show that $\mathcal{T} := \{t \geq 0 : \Delta x_{i,j}(t) < R_{\text{vis}}\} = [0, \infty)$ whenever $0 < \Delta x_{i,j}(0) < R_{\text{vis}}$ and $x_k(t) \notin (x_i(t), x_j(t))\ \forall k$ at any time $t \geq 0$. Suppose to the contrary that $\mathcal{T} \neq [0, \infty)$ and consider the leftmost connected component $[0, \tau), 0 < \tau < \infty$ of E. Then $\Delta(t) := \Delta x_{i,j}(t) < R_{\text{vis}}\ \forall t \in [0, \tau)$, whereas $\Delta(\tau) = R_{\text{vis}}$. Also, $d_j^-(t) = d_i^+(t) = \Delta(t)$ by (4), (5). With regard to Asm. 1, we have for a.a $t \in [0, \tau)$

$$\dot{\Delta}(t) = \dot{x}_j(t) - \dot{x}_i(t) \overset{(1),(6)}{=\!=} \Xi[\underbrace{d_j^+(t)}_{\leq R_{\text{vis}}}] - \Xi[d_j^-(t)] - \Xi[d_i^+(t)] + \Xi[\underbrace{d_i^-(t)}_{\leq R_{\text{vis}}}] \overset{(7)}{\leq} 2\Xi[R_{\text{vis}}] - 2\Xi[\Delta(t)]. \quad \text{(A.3)}$$

The associated ODE $\dot{\Delta} = 2\Xi[R_{\text{vis}}] - 2\Xi[\Delta]$ has a solution $\Delta_* \equiv R_{\text{vis}}$ such that $\Delta_*(0) > \Delta(0)$. By [32, Ch. III, Th. 4.1], $\Delta(\tau) < \Delta_*(\tau) = R_{\text{vis}}$, in violation of the above property $\Delta(\tau) = R_{\text{vis}}$. This contradiction completes the proof.

In the remainder of this appendix, we adopt the following.

Assumption A.2 *The initial state is collisionless and non-breaking.*

By Defn. 2.2, Cor. 3, and Lem. A.2, then the state is non-breaking for all $t \geq 0$. We invoke Cor. 2, enumerate the agents so that (2) holds, and put $\boldsymbol{\Delta}(t) := \left[\Delta_0(t), \ldots, \Delta_{N-2}(t)\right]^\top$.

Corollary 4 *The function $\boldsymbol{\Delta}(\cdot)$ is bounded on $[0, \infty)$.*

We recall that *ω-limit point* of $\boldsymbol{\Delta}(\cdot)$ is $\lim_{j\to\infty} \boldsymbol{\Delta}(t_j)$ associated with some unbounded sequence $t_1 < \ldots < t_j < \ldots$. By Cor. 4, the set Ω of all such points is nonempty and compact.

Lemma A.3 *The following inequality holds:*

$$\min_{\boldsymbol{\Delta} \in \Omega} \min_{i \in [0:N-2]} \Delta_i \geq w. \quad \text{(A.4)}$$

PROOF. Since $\Delta_i \geq 0$, (A.4) holds if $w = 0$. Let $w > 0$. Suppose that (A.4) fails to be true. We consider a minimizer $\boldsymbol{\Delta}$ for $\min_{\boldsymbol{\Delta}\in\Omega}$ in (A.4) and the related sequence $\{t_k\}_{k=1}^\infty$, denote $J := \{j \in [0 : N-2] : \Delta_j = \min_{i\in[0:N-2]} \Delta_i\} \neq \emptyset$ and put $i := \max_{j\in J} j, \Delta_{-1} := \Delta_{N-1} := w$. Then $\Delta_i < w, \Delta_{i+1} - \Delta_i > 0, \Delta_{i-1} \geq \Delta_i$. Due to (1) and (A.2), we have

$$|\dot{\Delta}_j| \leq 2\overline{u} \qquad \forall j \in [-1 : N-1]. \quad \text{(A.5)}$$

We pick $\xi > 0, \varepsilon > 0$ so that $\xi < \min\{\Delta_{i+1} - \Delta_i; w - \Delta_i\}, \varepsilon < \min\{\xi\mu_-/(2\mu_+); w - \xi - \Delta_i; \Delta_{i+1} - \Delta_i - \xi\}$, where

$$\mu_\pm := \pm \max_{z \in [0, R_{\text{vis}}]} \pm\Xi'(z) \overset{(7)}{>} 0, \quad \text{(A.6)}$$

and put $\delta := \varepsilon/(4\overline{u}), t_k^- := t_k - \delta$. For $t \in [t_k^-, t_k]$ and $k \approx \infty$,

$$\Delta_{i+1}(t) - \Delta_i(t) \overset{\text{(A.5)}}{\geq} \Delta_{i+1}(t_k) - \Delta_i(t_k) - 4\overline{u}\delta \geq \xi + \varepsilon - 4\overline{u}\delta = \xi,$$
$$\Delta_{i-1}(t) - \Delta_i(t) \overset{\text{(A.5)}}{\geq} \Delta_{i-1}(t_k) - \Delta_i(t_k) - 4\overline{u}\delta \geq -\varepsilon - 4\overline{u}\delta = -2\varepsilon.$$

We put $\varkappa := \mu_-\xi - 2\varepsilon\mu_+ > 0$. By Lem. 2, Cor. A.2, (4)–(7),

$$\dot{\Delta}_i(t) = \sum_{k=i-1,i} \{\Xi[\Delta_{k+1}(t)] - \Xi[\Delta_k(t)]\} \quad \text{(A.7)}$$
$$\geq \varkappa \Rightarrow \Delta_i(t_k^-) = \Delta_i(t_k) - \int_{t_k^-}^{t_k} \dot{\Delta}_i(t)\, dt \leq \Delta_i(t_k) - \delta\varkappa.$$

By picking a convergent subsequence from $\{\boldsymbol{\Delta}(t_k^-)\}_k$, we acquire $\boldsymbol{\Delta}^- \in \Omega$ with $\Delta_i^- \leq \Delta_i - \varkappa\delta$. Then $\min_{j\in[0:N-2]} \Delta_j^- < \Delta_i = \min_{i\in[0:N-2]} \Delta_j$. Thus $\boldsymbol{\Delta}$ is not a minimizer, in violation of its choice. This contradiction completes the proof.

Similar arguments justify the following.

Lemma A.4 $\max_{\boldsymbol{\Delta}\in\Omega} \max_{i\in[0:N-2]} \Delta_i \leq w$.

Lemma A.5 *For any $i \in [0 : N-2]$, we have $\Delta_i(t) \to w$ as $t \to \infty$, and this convergence is exponentially fast.*

PROOF. Lem. A.3, A.4 imply that Ω contains only one point $\boldsymbol{\Delta}_w$. This is that with $\Delta_i = w\ \forall i$. Thus all ω-limit points of the bounded function $\boldsymbol{\Delta}(\cdot)$ are the same and equal $\boldsymbol{\Delta}_w$. This implies that $\boldsymbol{\Delta}(t) \to \boldsymbol{\Delta}_w$ [32, Ch. VII, Cor. 1.1].

Thanks to the first inequality in (3) and the just justified convergence , $\Delta_i(t) < R_{\text{vis}}\ \forall i \in [0 : N-2]$ for all sufficiently large t's. We focus on them. Similarly to (A.3), we see that

$$\dot{\Delta}_i = \Xi[\Delta_{i+1}] + \Xi[\Delta_{i-1}] - 2\Xi[\Delta_i] \quad \forall i \in [0 : N-2]. \quad \text{(A.8)}$$

We use the change of the variable $\tau \mapsto \Xi'(w)t$ (where $\Xi'(w) > 0$ by (7)). The system (A.8) linearized at the equilibrium $\boldsymbol{\Delta_w}$

looks as follows $\dot{\mathbf{\Delta}} = (-2I + A)\mathbf{\Delta}$, where I is the $(N-1) \times (N-1)$ identity matrix and A is the square $(N-1)\times(N-1)$-matrix with 1's on the subdiagonal and superdiagonal and 0's in all other places. If $N = 2$, then $A = 0$ and so the spectral radius $\rho(A) = 0 < 2$. Let $N \geq 3$. The A-related graph $\Gamma(A)$ (defined in [33, Defn.6.2.11]) has the set of nodes $[0 : N-2]$, is undirected since A is symmetric, and has an edge that links nodes i and $i+1$ for any $i \in [0 : N-3]$. So $\Gamma(A)$ is strongly connected and hence the matrix A is irreducible by [33, Thm.6.2.24]. Insofar as its entries are nonnegative, $\rho(A)$ is its eigenvalue associated with an eigenvector $\mathbf{\Delta} = (\Delta_0, \ldots, \Delta_{N-2})^\top$ for which $\Delta_i > 0\ \forall i$ by [33, Thm.8.4.4]. So

$$A\mathbf{\Delta} = \rho(A)\mathbf{\Delta} \Leftrightarrow \Delta_1 = \rho(A)\Delta_0, \quad \Delta_0 + \Delta_2 = \rho(A)\Delta_1,$$
$$\Delta_1 + \Delta_3 = \rho(A)\Delta_2, \ldots, \Delta_{N-4} + \Delta_{N-2} = \rho(A)\Delta_{N-3},$$
$$\Delta_{N-3} = \rho(A)\Delta_{N-2}.$$

By summing up these equations, we see that

$$\rho(A)\sum_{i=0}^{N-2}\Delta_i = 2\sum_{i=0}^{N-2}\Delta_i - \Delta_0 - \Delta_{N-2} < 2\sum_{i=0}^{N-2}\Delta_i \Rightarrow \rho(A) < 2.$$

It follows that the linearized system $\dot{\mathbf{\Delta}} = (-2I + A)\mathbf{\Delta}$ is exponentially stable. The proof is completed by [34, Thm. 3.7].

Proof of Thm. 3.1: Claim **i)** is due Cor. 3 and Lem. A.2.

ii) Following **ii)**, we suppose that Asm. A.2 holds. Then

$$\dot{x}_0(t) \overset{(1)}{=} u_0(t) \overset{(6)}{=} \Xi[d_0^+(t)] - \Xi[d_0^-(t)] \xlongequal{(2),(4),(5)} \Xi[d_0^+(t)] - \Xi[w].$$

Here by (4), (5), (2), Lem. A.5, and the first inequality in (3), $d_0^+(t) = \Delta_0(t)$ for all sufficiently large t's. For them,

$$|\dot{x}_0(t)| = |\Xi[\Delta_0(t)] - \Xi[w]| \overset{(A.6)}{\leq} \mu_+|\Delta_0(0) - w|.$$

So by Lem. A.5, $\int_0^\infty |\dot{x}_0(t)|\, dt < \infty$ and $\exists x_* := \lim_{t\to\infty} x_0(t)$. For $i \in [1 : N-1]$, Lem. A.5 yields that $x_i(t) = x_0(t) + \sum_{j=0}^{i-1}\Delta_j(t) \xrightarrow{t\to\infty} x_* + iw$. By Defn. 2.1, this means that the agents do achieve an even self-deployment with the step w.

B Proof of Theorem 4.1

The cases 1) and 2) in Defn. 2.1 are handled concurrently, with giving remarks on case 2 when necessary. We recall that the anchor is labeled by 0.

Lemma B.1 *The claims* **i)** *and* **ii)** *of Thm.* 4.1 *hold.*

PROOF. ii) By the Danskin's theorem [31], the function $x_{\max}(t) := \max_i x_i(t)$ is absolutely continuous and $\dot{x}_{\max}(t) = \dot{x}_{i^{\max}(t)}(t)$ for a.a. t, where $i^{\max}(t)$ furnishes $\max_i x_i(t)$. If case 2 holds and $i^{\max}(t) = 0$, then $\dot{x}_{\max}(t) = \dot{x}_0(t) \equiv 0$. Otherwise, $\mathcal{I}^+_{i^{\max}(t)}(t) = \emptyset$ by (4), $d^+_{i^{\max}(t)}(t) = w = 0, d^-_{i^{\max}(t)}(t) \geq 0$ by (5), and so $\dot{x}_{\max}(t) \leq 0$ by (6) (where $i := i^{\max}(t)$), (1), and (7). So $\dot{x}_{\max}(t) \leq 0$ in any case and $x_{\max}(t)$ does not grow up. It remains to note that $x_{\min}(t) := \min_i x_i(t)$ does not decrease likewise.

i) It suffices to note that $\mathcal{R}(t) = x_{\max}(t) - x_{\min}(t)$.

Corollary 3 *There exist the limits* $m_+ := \lim_{t\to\infty} x_{\max}(t) \geq m_- := \lim_{t\to\infty} x_{\min}(t)$, *and they are finite.*

Let $y_1(\cdot), \ldots, y_M(\cdot) : [t_0, t_1] \to \mathbb{R}$ be absolutely continuous functions, and let $[k^1(t), \ldots, k^M(t)]$ be a permutation of $[1 : M]$ such that $z_1(t) := y_{k^1(t)}(t) \leq \ldots \leq z_M(t) := y_{k^M(t)}(t)$.

Lemma B.2 *For any i, the function $z_i(\cdot)$ is absolutely continuous and $\dot{z}_i(t) = \dot{x}_{k^i(t)}(t)$ for a.a. t.*

PROOF. We argue via induction on M. For $M = 1$, the claim trivially holds. Let it be true for some M and let $y_1(\cdot), \ldots, y_{M+1}(\cdot)$ be absolutely continuous functions. For the sequence $z_1(\cdot), \ldots, z_M(\cdot)$ fabricated of $y_1(\cdot), \ldots, y_M(\cdot)$, the claim is true by the induction hypothesis. For $\nu = 0, \ldots, M$, we define y^ν_{M+1} via the recursion:

$$y^0_{M+1}(t) := y_{M+1}(t), \quad y^\nu_{M+1}(t) := \max\{z_\nu(t); y^{\nu-1}_{M+1}(t)\},$$
$$\widehat{z}_\nu(t) := \min\{z_\nu(t); y^{\nu-1}_{M+1}(t)\}.$$

Then $\widehat{z}_1(t) \leq \ldots \leq \widehat{z}_M(t) \leq \widehat{z}_{M+1}(t) := y^M_{M+1}(t)$ results from $y_1(t), \ldots, y_{M+1}(t)$ via a permutation of the indices. By consecutively applying the Danskin's theorem [31] at every step of the recursion, we see that the functions $\widehat{z}_\nu(\cdot)$ are absolutely continuous. For any ν, the sequence

$$\mathcal{Z}_\nu(t) = [\widehat{z}_1(t), \ldots, \widehat{z}_\nu(t), z_{\nu+1}(t), \ldots, z_M(t), y^\nu_{M+1}(t)] \quad \text{(B.1)}$$

results from $\mathcal{Z}_{\nu-1}$ by applying a permutation $J_\nu(t)$ of the indices, which either swaps the νth and $(M+1)$th elements or maps the element into itself. Let $\mathcal{Z}'_\nu(t)$ be obtained from (B.1) via replacing every function by its derivative. By the Danskin's theorem [31], $\mathcal{Z}'_\nu(t)$ results from $\mathcal{Z}'_{\nu-1}(t)$ by applying the same permutation $J_\nu(t)$ of the indices. The sequence $\mathcal{Z}_0(t)$ results from $\mathcal{Y}(t) := [y_1(t), \ldots, y_{M+1}(t)]$ by applying a permutation of the form $\mathcal{K}(t) = [k^1(t), \ldots, k^M(t), M+1]$. By the induction hypothesis, $\mathcal{K}(t)$ also relates $\mathcal{Z}'_0(t)$ and $\mathcal{Y}'(t) := [y'_1(t), \ldots, y'_{M+1}(t)]$. The proof is completed by noting that $\mathcal{K}_{M+1}(t) = J_M \circ \cdots \circ J_1 \circ \mathcal{K}(t)$ maps $\mathcal{Y}'(t)$ into $\mathcal{Z}'_M(t)$.

Let $\{x_i(t)\}_{i=0}^{N-1}, t \geq 0$ be a trajectory under the protocol $1_{\max}$, and let $[k^0(t), \ldots, k^{N-1}(t)]$ be a permutation of $\{0, \ldots, N-1\}$ such that $z_0(t) := x_{k^0(t)}(t) \leq \ldots \leq z_{N-1}(t) := x_{k^{N-1}(t)}(t)$. For any time-dependent event $\mathcal{E}$, we put $I_{\mathcal{E}}(t) := 1$ if this event occurs at time t, and $I_{\mathcal{E}}(t) := 0$ otherwise. Finally, $I^{\neq a}_s(t) := 1$ in the case 1 and $I^{\neq a}_s(t) := I_{k^s > 0}(t)$ in the case 2. The event symbolized by $i \sim j$ means that $k^i(t)$ and $k^j(t)$ are linked in $\Gamma_{\text{all}}(t)$.

We see that by (1), (7), (6), and (5) (with min $\mapsto$ max),

$$\begin{aligned} d^-_{k^s(t)}(t) &= \max_{i=0,\dots,s-1} \left\{[x_{k^s(t)}(t) - x_{k^i(t)}(t)]I_{s\sim i}(t)\right\} \\ &= \max_{i=0,\dots,s-1} \{[z_s(t) - z_i(t)]I_{s\sim i}(t)\}\,; \\ d^+_{k^s(t)}(t) &= \max_{i=s+1,\dots,N-1} \{[z_i(t) - z_s(t)]I_{s\sim i}(t)\}\,; \\ \dot z_s(t) &= I^{\neq a}_s(t) \max_{i>s} \Xi[z_i(t) - z_s(t)]I_{s\sim i}(t) \\ &\quad - I^{\neq a}_s(t) \max_{i<s} \Xi[z_s(t) - z_i(t)]I_{s\sim i}(t). \end{aligned} \tag{B.2}$$

We put $g_s(t) := \min\{z_s(t), x_\dagger\}$($x_\dagger := \infty$ in the case 1). Then by the Danskin's theorem [31]

$$\begin{aligned} \dot g_s(t) &= I_{z_s<x_\dagger}(t)\dot z_s(t) \\ &= \max_{i>s} \Xi[z_i(t) - z_s(t)]I_{s\sim i}(t)I^{\neq a}_s(t)I_{z_s<x_\dagger}(t) \\ &\quad - \max_{i<s} \Xi[z_s(t) - z_i(t)]I_{s\sim i}(t)I^{\neq a}_s(t)I_{z_s<x_\dagger}(t). \end{aligned} \tag{B.3}$$

Lemma B.3 *For any $i < j$, the integral $\Sigma_{i\to j} := \int_0^\infty \Xi[z_j(t) - z_i(t)]I_{i\sim j}(t)I_{x_\dagger>z_j}(t)\, dt < \infty$ is finite.*

PROOF. We show that this holds whenever $i \le p$ and for all $p \in [0 : N-1]$, arguing via induction on p. If $p = 0$, then $i = 0$, whereas (B.3) with $s := 0$ implies that for any $j > 0$,

$$\begin{aligned} \dot z_0(t) &\ge \Xi[z_j(t) - z_0(t)]I_{0\sim j}(t)I^{\neq a}_s(t) \\ &\ge \Xi[z_j(t) - z_0(t)]I_{0\sim j}(t)I_{z_j<x_\dagger}(t) \\ \Rightarrow \Sigma_{0,j} &\le \varlimsup_{t\to\infty} z_0(t) - z_0(0) = m_- - z_0(0) \overset{\text{Cor. 3}}{<} \infty. \end{aligned}$$

Let $\Sigma_{i\to j} < \infty\ \forall i < j$ if $i \le p \le N-2$, and consider i, j such that $i < j$ and $i \le p+1$. If $i \le p$, then $\Sigma_{i\to j} < \infty$ by the induction hypothesis. Let $i = p+1$. By (B.3) with $s := i$

$$\begin{aligned} \dot g_i(t) = I_{z_i<x_\dagger}(t)\dot z_i(t) &\ge \Xi[z_j(t) - z_i(t)]I_{i\sim j}(t)I_{z_j<x_\dagger}(t)I_{z_i<x_\dagger}(t) \\ &\quad - \sum_{r\le p} \Xi[z_i(t) - z_r(t)]I_{i\sim r}(t)I^{\neq a}_i(t)I_{z_i<x_\dagger}(t). \end{aligned}$$

Here $I_{z_j<x_\dagger}(t) = 1 \xRightarrow{z_i(t)\le z_j(t)} I_{z_i<x_\dagger}(t) = 1 \Rightarrow I_{z_j<x_\dagger}(t) = I_{z_j<x_\dagger}(t)I_{z_i<x_\dagger}(t)$. Since $I^{\neq a}_i(t) \le 1$, we see that

$$\begin{aligned} \dot g_i(t) &\ge \Xi[z_j(t) - z_i(t)]I_{i\sim j}(t)I_{z_j<x_\dagger}(t) \\ &\quad - \sum_{r\le p} \Xi[z_i(t) - z_r(t)]I_{i\sim r}(t)I_{z_i<x_\dagger}(t) \\ \Rightarrow \int_0^\infty \Xi[z_j(t) - z_i(t)]&I_{i\sim j}(t)I_{z_j<x_\dagger}(t)\, dt \le \varlimsup_{t\to\infty} g_i(t) - g_i(0) \\ &\quad + \sum_{r\le p} \int_0^\infty \Xi[z_i(t) - z_r(t)]I_{i\sim r}(t)I_{z_i<x_\dagger}(t)\, dt. \end{aligned}$$

It remains to note that $i = p+1$ and so every integral in $\sum_{i\le r} \dots$ is finite by the induction hypothesis.

Lemma B.4 $\int_0^\infty |\dot g_s(t)|\, dt < \infty$ *and* $\lim_{t\to\infty} g_s(t) \in \mathbb{R}$ *exists.*

PROOF. We note that the former implies the latter.

Case 1 in Defn. 2.1: It suffices to note that by (B.3) and Lem. B.3 (with $x_\dagger = \infty$), $\int_0^\infty |\dot g_s(t)|dt = \int_0^\infty |\dot z_s(t)|dt < \infty$.

Case 2 in Defn. 2.1: By (B.3), $g_s(\cdot) = g_{s,1}(\cdot) - g_{s,2}(\cdot)$, where $g_{s,1}(\cdot)$ and $g_{s,2}(\cdot)$ are the antiderivatives of the minuend and subtrahend, respectively, in (B.3). By (7), $z_i(t) \ge z_s(t)\ \forall i > s \Rightarrow \dot g_{s,1}(t) \ge 0$ a.a. t, and so $\exists \lim_{t\to\infty} g_{s,1}(t) \in (-\infty, +\infty]$. Also, $|\dot g_{s,2}(t)| \le \sum_{i<s} \Xi[z_s(t) - z_i(t)]I_{s\sim i}(t)I_{z_s<x_\dagger}(t)$ and so $\int_0^\infty |\dot g_{s,2}(t)|\, dt < \infty$ by Lem. B.3. Hence $\exists \lim_{t\to\infty} g_{s,2}(t) \in (-\infty, +\infty)$, and $g^\infty := \lim_{t\to\infty} g_s(t) \in (-\infty, +\infty]$ does exist. Here $g_s(t) \le x_\dagger\ \forall t \Rightarrow g^\infty \neq +\infty$ and so $\int_0^\infty |\dot g_{s,1}(t)|\, dt < \infty \Rightarrow \int_0^\infty |\dot g_s(t)|\, dt < \infty$.

By arguing like when proving Lem. B.3, B.4 (but using the backward induction on p), we arrive at the following fact about $h_s(t) := \max\{z_s(t), x_\dagger\}$.

Lemma B.5 *In the case 2, the limit $h_s^\infty := \lim_{t\to\infty} h_s(t) \in \mathbb{R}$ exists and $\int_0^\infty |\dot h_s(t)|\, dt < \infty$ for any s.*

Since either $g_s^\infty < x_\dagger$, or $h_s^\infty > x_\dagger$, or $h_s^\infty = g_s^\infty = x_\dagger$ in the case 2, and $g_s(\cdot) \equiv z_s(\cdot)$ in the case 1, the following holds.

Corollary 4 *There exists the limit $z_s^\infty := \lim_{t\to\infty} z_s(t) \in \mathbb{R}$.*

Proof of Thm. 4.1: i) and ii) are established by Lem. B.1.

iii) By Cor. 3 and 4, $\min_s z_s^\infty = m_- \le m_+ = \max_s z_s^\infty$, where $m_- \le x_\dagger \le m_+$ in the case 2. It suffices to show that $z_s^\infty = m_-\ \forall s$. For $s = 0$, this is true since $z_0(\cdot) \equiv x_{\min}(\cdot)$. Suppose that this is untrue for some s, and consider the least such s. Then $s > 0$, $z_{s-1}^\infty = m_- < z_s^\infty$ and so there is $t_* \ge 0$, $\varepsilon > 0$ such that $z_{s-1}(t) \le m_- + \varepsilon < m_- + 2\varepsilon < z_s(t)\ \forall t \ge t_*$.

We put $G_- := \{k_0(t_*), \dots, k_{s-1}(t_*)\}$ and $G_+ := \{k_s(t_*), \dots, k_{N-1}(t_*)\}$. For $t \ge t_*$, the inequality $z_{s-1}(t) < z_s(t)$ implies that the chains $\{k_0(t), \dots, k_{s-1}(t)\}$ and $\{k_s(t), \dots, k_{N-1}(t)\}$ are permutations of G_- and G_+, respectively. The event symbolized by $G_- \sim G_+$ means that $t \ge t_*$ and the graph $\Gamma_*(t)$ contains an edge bridging G_- and G_+. Then $I_{G_-\sim G_+}(t) = 1 \Rightarrow \exists i < s, j \ge s : I_{i\sim j}(t) = 1$ and $z_j(t) - z_i(t) \ge z_s(t) - z_{s-1}(t) \ge \varepsilon \Rightarrow \Xi[z_j(t) - z_i(t)] \ge \Xi[\varepsilon] > 0$. Hence

$$\begin{aligned} \Xi[\varepsilon] \int_{t_*}^\infty I_{G_-\sim G_+}(t)\, dt \quad & \\ \le \sum_{i<s, j\ge s} \int_{t_*}^\infty I_{i\sim j, \Xi[z_j(t)-z_i(t)]\ge\Xi[\varepsilon]}(t)\Xi[\varepsilon]\, dt & \\ \le \sum_{i<s, j\ge s} \int_{t_*}^\infty I_{i\sim j}(t)\Xi[z_j(t) - z_i(t)]\, dt \overset{\text{in case 1 by Lem. B.3}}{\underset{x_\dagger=\infty \text{ in case 1}}{<}} \infty. & \end{aligned} \tag{B.4}$$

Hence G_- and G_+ are not bridged in Γ_{ess}, in violation of an assumption of Thm. 4.1. This proves Thm. 4.1 in the case 1.

Case 2. The anchor $i = 0$ is in $G_- \cup G_+$. Let $0 \in G_+$, the case $0 \in G_-$ is handled likewise. Then for $i < s, t \ge t_*$, we have $z_i(t) \le z_{s-1}(t) < z_s(t) \le x_\dagger$ and by Lem. B.3 $\int_0^\infty \Xi[z_j(t) -$

$z_i(t)]I_{i\sim j}(t)\ dt = \int_0^\infty \Xi[z_j(t)-z_i(t)]I_{i\sim j}(t)I_{x_\dagger>z_j}(t)\ dt = \Sigma_{i\to j} < \infty\ \forall i<j<s$. By (B.2) (with $s:=i$),

$$\begin{aligned} z_i^\infty - z_i(t_*) = \int_{t_*}^\infty \dot z_i(t)\ dt = \int_{t_*}^\infty \max_{j>i} \Xi[z_j(t)-z_i(t)]I_{i\sim j}(t)\ dt \\ - \int_{t_*}^\infty \max_{j<i} \Xi[z_i(t)-z_j(t)]I_{i\sim j}(t)\ dt \\ \geq \int_{t_*}^\infty \max_{j>i} \Xi[z_j(t)-z_i(t)]I_{i\sim j}(t)\ dt - \sum_{j=0}^{i-1} \Sigma_{j,i}. \end{aligned}$$

Here any addend $\Sigma_{j,i} < \infty$ and so $\int_{t_*}^\infty \max_{j>i}\Xi[z_j(t) - z_i(t)]I_{i\sim j}(t)\ dt < \infty \Rightarrow \int_{t_*}^\infty \Xi[z_j(t)-z_i(t)]I_{i\sim j}(t)\ dt < \infty\ \forall j>i$. Hence the last inequality in (B.4) still holds. The proof is then completed like in the case 1. □

C Proofs of Theorems 5.1 and 5.2.

The contexts of Thm. 5.1 and 5.2 are called the *case* 1 and 2, respectively. We alter (A.2): $\Delta_{-1}(t) := x_0(t)-x_\dagger$ in the case 1 and $\Delta_{-1}(t) := x_0(t)-x_-, \Delta_{N-1}(t) := x_+ - x_{N-1}(t)$ in the case 2. We put $x_{-1}(t) := x_\dagger, x_N(t) := \infty$ in the case 1 and $x_{-1}(t) := x_-, x_N(t) := x_+$ in the case 2. Then the first formula in (A.2) extends on $i \in [-1 : N-2]$ in the case 1 and on $i\in[-1:N-1]$ in the case 2. [1] Formula (2) extends into

$$x_{-1}(t) < x_0(t) < x_1(t) < \ldots < x_{N-1}(t) < x_N(t) \forall t\geq 0. \quad \text{(C.1)}$$

Lemma C.1 *Lem.* A.1 *remains true for* $i,j\in[-1:N]$.

PROOF. It suffices to retrace the proof of Lem. A.1 with complementing (A.1). The last is needed in the situation where some of the "agents" $i^+(t), i^-(t)$ is an anchor in the case 1 or an end of $[x_-,x_+]$ in the case 2. Since $x_{i^+(t)}(t) > x_{i^-(t)}(t)$ and $x_{i^+(t)}(t) \approx x_{i^-(t)}(t)$ (for $t<\tau, t\approx\tau$), only one of the "agents" $i^+(t), i^-(t)$ is an anchor in the case 1 or an end of the interval in the case 2. Then in (A.1), the related derivative $\dot x_{i^\pm(t)}(t)$ is dropped, along with its descendants. This alters the outcome $\dot\Delta(t) \leq -2\Xi(\Delta)$ of (A.1) into $\dot\Delta(t)\leq -\Xi(\Delta)$. It remains to note that the concluding part of the proof easily survives this change of the multiplier $2\mapsto 1$.

Corollary 5 *Let Asm.* A.1 *be true. Then* (C.1) *holds.*

Corollary 6 *Both protocols* 2 *and* 3 *are collisionless.*

Lemma C.2 *Protocols* 2 *and* 3 *are connectivity saving.*

PROOF. It suffices to retrace the proof of Lem. A.2, while complementing it, where necessary, with observations and modifications similar to those from the proof of Lem. C.1.

Remark C.1 *The same arguments show that in the case 1,* $\Delta_{-1}(0) < R_{\text{vis}} \Rightarrow \Delta_{-1}(t) < R_{\text{vis}}\ \forall t\geq 0$.

[1] For **4** in Sect. 6, the r.h.s. of all formulas for Δ_{-1} and Δ_{N-1} should be multiplied by α.

Proof of Thm. 5.2: By Cor. 6 and Lem. C.2, **i)** in Thm. 3.1 is true. Let Asm. A.1 hold. Since $x_{-1}(t)\equiv x_-, x_N(t)\equiv x_+$ in (C.1), the state $X(t) := \{x_i(t)\}_{i=0}^{N-1}$ is bounded on $[0,\infty)$. Let $\Omega_X \subset [x_-,x_+]$ stand for the compact and nonempty set of all its ω-limit points; by (C.1), $x_{i+1}\geq x_i\ \forall i\in[0:N-2], X=\{x_i\}_{i=0}^{N-1}\in\Omega_X$. We intend to show that

$$\min_{X\in\Omega_X}\min_{j\in[-1:N-1]} \Delta_j(X) \geq q := \frac{L}{N+1}, \qquad \text{(C.2)}$$

where $L := x_+ - x_-$. Suppose the contrary and take a minimizer $X\in\Omega_X$ of $\min_{X\in\Omega_X}$ and the related sequence $\{t_k\}_{k=1}^\infty$. For any minimizer i of $\min_{j\in[-1:N-1]}$ from (C.2), $\Delta_i = x_{i+1}-x_i < q$. Since $\sum_{i=-1}^{N-1}\Delta_i = L$, there is $j\in[-1:N-1]$ such that $\Delta_j>q$. Hence i can be chosen so that either **1)** $i\geq 0$ and $\Delta_{i-1}>q$ or **2)** $i\leq N-2$ and $\Delta_{i+1}>q$. We focus on **1)**; the situation **2)** is treated likewise.

By the assumptions of Thm. 5.2, $q<w<R_{\text{vis}}$. We take $\mu_\pm$ from (A.6) and pick $\xi\in\big(0,\min\{\Delta_{i-1}-\Delta_i; w-q\}\big)$, $0<\varepsilon<\min\left\{\frac{\xi\mu_-}{2\mu_+}; w-q-\xi; \Delta_{i-1}-\Delta_i-\xi\right\}$, $\delta := \varepsilon/(4\overline{u})$, $t_k^- := t_k-\delta$. Then for $t\in[t_k^-,t_k]$ and sufficiently large k's, we have

$$\begin{aligned} \Delta_i(t)+\xi \leq \Delta_i(t_k)+\xi+2\overline{u}\delta \leq q+\xi+\varepsilon\leq w < R_{\text{vis}}, \\ \Delta_{i-1}(t)-\Delta_i(t) \geq \Delta_{i-1}(t_k)-\Delta_i(t_k)-4\overline{u}\delta \geq \xi+\varepsilon-4\overline{u}\delta\geq\xi, \\ \Delta_{i+1}(t)-\Delta_i(t)\geq\Delta_{i+1}(t_k)-\Delta_i(t_k)-4\overline{u}\delta\geq -\varepsilon-4\overline{u}\delta\geq-2\varepsilon. \end{aligned} \qquad \text{(C.3)}$$

By (5), $d_i^-(t)\geq\min\{\Delta_{i-1}(t);w\}\geq\Delta_i(t)+\xi$ and if $i\leq N-2$, then $d_{i+1}^+(t)\geq\min\{\Delta_{i+1}(t);w\}\geq\Delta_i(t)-2\varepsilon$. So if $i\leq N-2$,

$$\begin{aligned} \dot\Delta_i(t) = \Xi[d_{i+1}^+(t)]+\Xi[d_i^-(t)]-2\Xi[\Delta_i(t)] \overset{(7)}{\geq} \Xi[\Delta_i(t)-2\varepsilon] \\ -\Xi[\Delta_i(t)]+\Xi[\Delta_i(t)+\xi]-\Xi[\Delta_i(t)] \overset{(A.6)}{\geq} \mu_-\xi-2\mu_+\varepsilon. \end{aligned}$$

If $i=N-1$, then $\dot x_{i+1}=\dot x_N = 0$ and, as above,

$$\begin{aligned} \dot\Delta_i(t) = -\dot x_i(t) = \Xi[d_i^-(t)]-\Xi[\Delta_i(t)]\geq\mu_-\xi \\ > \varkappa := \mu_-\xi-2\mu_+\varepsilon > 0. \end{aligned}$$

Anyway, $\Delta_i(t_k^-) = \Delta_i(t_k)-\int_{t_k^-}^{t_k}\dot\Delta_i(t)\ dt\leq\Delta_i(t_k)-\delta\varkappa$. By picking a convergent subsequence from $\{X(t_k^-)\}_k$, we get an ω-limit point $X^\downarrow$ with $\Delta_i^\downarrow\leq\Delta_i-\varkappa\delta<\min_{j\in[-1:N-1]}(x_{j+1}-x_j)$, in violation of the definition of X as a minimizer in (C.2). This contradiction proves the inequality (C.2).

For any $X\in\Omega_X$, we have $x_{i+1}-x_i\geq\frac{L}{(N+1)}\forall i\in[-1:N-1]$ by (C.2), and $\sum_{i=-1}^{N-1}(x_{i+1}-x_i) = L$. So $x_{i+1}-x_i = q\ \forall i\in[-1:N-1]$ and $x_{-1}=x_-\Rightarrow x_i = x_-+iq\ \forall i\in[-1:N]$ for any ω-limit point X. Thus, there is only one such a point, and so $x_i(t)\to x_-+iq$ as $t\to\infty$ for all $i\in[0:N-1]$ [32, Ch. VII, Cor. 1.1]. Defn. 2.1 completes the proof. □

From now on, we focus on Thm. 5.1 and **adopt Asm. A.1**.

Lemma C.3 *If* $i\in[-1:N-2]$ *and* $\Delta_i(t)\geq R_{\text{vis}}$*, then* $\dot x_{i+1}(t)\leq 0$.

PROOF. Let $\Delta_i(t) \geq R_{\text{vis}}$. By the definition of protocol 2, $d^-_{i+1}(t) = R_{\text{vis}}$, whereas $d^+_{i+1}(t) \leq R_{\text{vis}}$ by Asm. 1, (4), (5), and the first part of (3). So by (1), (7), and (6), $\dot{x}_{i+1}(t) = u_{i+1}(t) = \Xi[d^+_{i+1}(t)] - \Xi[d^-_{i+1}(t)] \leq 0$.

Lemma C.4 *$X(t) := \{x_i(t)\}_{i=0}^{N-1}$ is bounded on $[0,\infty)$.*

PROOF. It suffices to show that $x_i(\cdot)$ is bounded by using induction on $i \in [-1 : N-1]$. Since $x_{-1}(t) \equiv x_\dagger$, the claim is true for $i = -1$. Let it hold for some $i \in [-1 : N-2]$, i.e., $b := \sup_{t\geq 0} x_i(t) < \infty$. If **i)** $\Delta_i(t) \geq R_{\text{vis}}\ \forall t \geq 0$, then $\dot{x}_{i+1}(t) \leq 0\ \forall t \geq 0$ by Lem. C.3 and so $x_{i+1}(t) \leq x_{i+1}(0) < \infty\ \forall t \geq 0$. If **i)** does not hold, there is t_* such that $\Delta_i(t_*) < R_{\text{vis}}$. Then $t \geq t_* \Rightarrow \Delta_i(t) < R_{\text{vis}}$ by Lem. C.2 and so $x_{i+1}(t) = x_i(t) + \Delta_i(t) \leq b + R_{\text{vis}}$. Hence $\sup_{t\geq 0} x_{i+1}(t) \leq \max\left\{ \max_{t\in[0,t_*]} x_{i+1}(t); b + R_{\text{vis}} \right\} < \infty$.

By Lem. C.4, the set Ω_X of ω-limit points of $X(\cdot)$ is compact and $\Omega_X \neq \emptyset$. By Asm. A.1 and Lem. 6, $x_{i+1} \geq x_i \forall i, X \in \Omega_X$.

Lemma C.5 *$\Delta_i < R_{\text{vis}}\ \forall i \in [-1 : N-1]$ for any $X \in \Omega_X$.*

PROOF. We argue via induction on $i = N-1, \ldots, -1$. For $i = N-1$, the claim follows from the first relation in (3) since $\Delta_{N-1} \equiv w$. Let the claim hold for $i \in [0 : N-1]$, i.e., $\zeta := \max_{X\in\Omega_X} \Delta_i < R_{\text{vis}}$. We pick $\xi \in (\max\{\zeta; w\}, R_{\text{vis}})$. Then $\Delta_i(t) < \xi\ \forall t \approx \infty$ since otherwise, $\Delta_i(t_k) \geq \xi\ \forall k$ for some sequence $t_1 < t_2 < \ldots < t_k \xrightarrow{k\to\infty} \infty$, and by picking a convergent subsequence $\{X(t_{k_j})\}_j$ from the bounded sequence $\{X(t_k)\}_k$ and taking its limit $X \in \Omega_X$, we arrive at the contradiction $\Delta_i(X) \geq \xi > \zeta$ with the definition of ζ.

Let us show that $\Delta_{i-1}(t) < R_{\text{vis}}\ \forall t \approx \infty$. By Lem. C.2 and Rem. C.1, $\Delta_{i-1}(\tau) < R_{\text{vis}} \Rightarrow \Delta_{i-1}(t) < R_{\text{vis}}\ \forall t \geq \tau$. So it suffices to show that $\exists \tau \geq 0 : \Delta_{i-1}(\tau) < R_{\text{vis}}$. Suppose to the contrary that $\Delta_{i-1}(t) \geq R_{\text{vis}}\ \forall t \geq 0$. By Asm. 1, (4), (5), and the definition of protocol 2, $d^-_i(t) = R_{\text{vis}}$ and $d^+_i(t) \leq \xi\ \forall t \approx \infty$. So (1), (6), (7), $\Rightarrow \dot{x}_i = u_i = \Xi[d^+_i] - \Xi[d^-_i] \leq \Xi[w] - \Xi[R_{\text{vis}}] < 0\ \forall t \approx \infty \Rightarrow x_i(t) \to -\infty$ as $t \to \infty$, in violation of (C.1), where $x_{-1} \equiv x_\dagger$. Thus $\Delta_{i-1}(t) < R_{\text{vis}}\ \forall t \geq \tau$ if $\tau \approx \infty$.

We focus on $t \geq \tau$ with such τ. Let $i \geq 1$. By (1), (4)–(6),

$$\dot{\Delta}_{i-1} = \Xi[d^+_i] + \Xi[d^-_{i-1}] - 2\Xi[\Delta_{i-1}] \leq \Xi[\xi] + \Xi[R_{\text{vis}}] - 2\Xi[\Delta_{i-1}]. \tag{C.4}$$

By [32, Ch. III, Th. 4.1], $\Delta_{i-1}(t) \leq \Delta_*(t)\ \forall t \geq \tau$, where $\Delta_*(\cdot)$ solves ODE $\dot{\Delta}_* = \Xi[\xi] + \Xi[R_{\text{vis}}] - 2\Xi[\Delta_*]$ and $\Delta_*(\tau) = \Delta_i(\tau) \in (0, R_{\text{vis}})$. By (7), $\frac{1}{2}[\Xi(\xi) + \Xi(R_{\text{vis}})] \in \big(\Xi(0), \Xi(R_{\text{vis}})\big) \Rightarrow \exists \lambda \in (0, R_{\text{vis}})$ such that $\Xi(\lambda) = \frac{1}{2}[\Xi(\xi) + \Xi(R_{\text{vis}})]$. Then $\dot{\Delta}_* = g(\Delta_*)$, where $g(z) := 2[\Xi(\lambda) - \Xi(z)], g(\lambda) = 0$, and $g(z)(z-\lambda) < 0\ \forall z \in (0, R_{\text{vis}}), z \neq \lambda$. Hence the constant solution $\Delta(t) \equiv \lambda$ of ODE $\dot{z} = g(z)$ attracts all solutions starting in $(0, R_{\text{vis}})$. In particular, $\Delta_*(t) \to \lambda$ as $t \to \infty$ and so $\overline{\lim}_{t\to\infty} \Delta_{i-1}(t) \leq \lambda$. It follows that $\Delta_{i-1} \leq \lambda < R_{\text{vis}}\ \forall X \in \Omega_X$. If $i = 0$, (C.4) alters into $\dot{\Delta}_{-1} = \Xi[d^+_0] - \Xi[\Delta_{-1}] \leq \Xi[\xi] - \Xi[\Delta_{-1}]$. As above, this entails that $\overline{\lim_{t\to\infty}} \Delta_{-1}(t) \leq \xi \Rightarrow \Delta_{-1} \leq \xi < R_{\text{vis}}\ \forall X \in \Omega_X$.

Lemma C.6 $\max_{\boldsymbol{X}\in\Omega_X} \max_{i\in[-1:N-2]} \Delta_i \leq w$.

PROOF. Suppose the contrary and consider a maximizer $X \in \Omega_X$ of $\max_{\boldsymbol{X}\in\Omega_X}$ and the related sequence $\{t_k\}_{k=1}^\infty$. For any maximizer $i \in [-1 : N-2]$ of the latter maximum, $\Delta_i = x_{i+1} - x_i > w$, whereas $\Delta_{N-1} = w$. So there is a maximizer $i \in [-1 : N-2]$ such that $\Delta_i > \Delta_{i+1}$. With $\mu_\pm$ from (A.6) at hands, we pick $\xi \in \big(0, \min\{\Delta_i - \Delta_{i+1}; R_{\text{vis}} - w\}\big)$, $0 < \varepsilon < \min\{\xi\mu_-/(2\mu_+); \Delta_i - \Delta_{i+1} - \xi\}$, $\delta := \varepsilon/(4\overline{u})$, $\quad t^-_k := t_k - \delta$. Then for $t \in [t^-_k, t_k]$ and sufficiently large k's, we have, like in (C.3) and with regard to Lem. C.5, we have $\Delta_j(t) < R_{\text{vis}}\ \forall j$ and

$$\Delta_i(t) - \Delta_{i+1}(t) \geq \Delta_i(t_k) - \Delta_{i+1}(t_k) - 4\overline{u}\delta \geq \xi + \varepsilon - 4\overline{u}\delta = \xi,$$
$$i \geq 0 \Rightarrow \Delta_i(t) - \Delta_{i-1}(t) \geq \Delta_i(t_k) - \Delta_{i-1}(t_k) - 4\overline{u}\delta$$
$$\geq -\varepsilon - 4\overline{u}\delta = -2\varepsilon.$$

By Asm. 1 and (4), (5), $d^+_{i+1}(t) = \Delta_{i+1}(t), d^-_{i+1}(t) = \Delta_i(t) = d^+_i(t)$ and $i \geq 0 \Rightarrow d^-_i(t) = \Delta_{i-1}(t)$. So for $i \geq 0$, (A.7) still holds by (1) and (6), and so

$$\dot{\Delta}_i(t) = \Xi[\Delta_{i+1}(t)] + \Xi[\Delta_{i-1}(t)] - 2\Xi[\Delta_i(t)] \leq -\varkappa$$
$$:= -\mu_-\xi + 2\varepsilon\mu_+ < 0$$
$$\Rightarrow \Delta_i(t^-_k) = \Delta_i(t_k) - \int_{t^-_k}^{t_k} \dot{\Delta}_i(t)\,dt \geq \Delta_i(t_k) + \delta\varkappa.$$

By picking a convergent subsequence from the bounded sequence $\{X(t^-_k)\}_k$, we get $X^- \in \Omega_X$ with $\Delta^-_i \geq \Delta_i + \varkappa\delta$. Then $\max_{i\in[-1:N-2]} \Delta^-_j \geq \Delta^-_i > \Delta_i = \max_{i\in[-1:N-2]} \Delta_j$, in violation of the definition of X as a maximizer. This contradiction completes the proof if $i \geq 0$.

If $i = -1$, (A.7) alters into $\dot{\Delta}_{-1}(t) = \Xi[\Delta_0(t)] - \Xi[\Delta_{-1}(t)] \leq -\mu_-\xi < 0$. So up to secondary details, the just used line of reasoning remains valid and leads to completing the proof.

Lemma C.7 $\min_{\boldsymbol{X}\in\Omega_X} \min_{i\in[-1:N-2]} \Delta_i \geq w$.

PROOF. It suffices to retrace the proof of Lem. A.3, while putting $[-1 : N-2]$ in place of $[0 : N-2]$ and dropping $\Delta_{-1} := w$.

Proof of Thm. 5.1: By Cor. 6 and Lem. C.2, **i)** in Thm. 3.1 is true. Let Asm. A.2 holds. Then for any $X \in \Omega_X$, Lem. C.6 and C.7 guarantee that $x_{i+1} - x_i = w\ \forall i \in [-1 : N-2]$, where $x_{-1} = x_\dagger$, and so $x_i = x_\dagger + (i+1)\ \forall i \in [0 : N-1]$. Thus there is only one ω-limit point, and so $x_i(t) \to x_\dagger + w(i+1)$ as $t \to \infty$ for all $i \in [0 : N-1]$ [32, Ch. VII, Cor. 1.1]. Defn. 2.1 completes the proof.

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